GOVERNMENT POLYTECHNIC AWASARI (KHURD)
Tal: Ambegaon, Pune - 412 405
2024-2025

Micro Project Report
On

"E-Learning Platform"

of Subject
Advanced Java Programming

Submitted in fulfillment for 'I' Scheme Fifth semester of
Diploma in
COMPUTER ENGINEERING

Submitted By:

Sr No	Roll No	Name Of Student	Enrollment No
1	23CO349	Kadambari Hemant Nagare	2210510059
2	23CO350	Aditi Vijay Jadhav	2210510060
3	23CO351	Aditya Babasaheb Dhanwade	2210510061

Under the Guidance of
Smt.Ajita.S.Patil

Maharashtra State Board of Technical Education, Mumbai
(Autonomous) (ISO-9001-2008) (ISO/IEC 27001:2013)

CERTIFICATE OF COMPLETION

This is to certify that the following students of Fifth Semester of Diploma in Computer Engineering of institute Government Polytechnic Awasari(Kh) (1051) have successfully completed microproject **"E-Learning Platform"** in the course of **"Advanced Java Programming" (22517)** for the academic year 2024-25.

Sr No	Roll No	Name Of Student	Enrollment No
1	23CO349	Kadambari Hemant Nagare	2210510059
2	23CO350	Aditi Vijay Jadhav	2210510060
3	23CO351	Aditya Babasaheb Dhanwade	2210510061

Place: Awasari(Kh)

Date: / /

Guide	HOD	Principal
(Smt.Ajita.S.Patil)	(Mrs.Megha.G.Yawalkar)	(Dr. Vitthal.S.Bandal)

ACKNOWLEDGEMENT

We have great pleasure and sense of satisfaction in presenting this micro-project report on **"E-Learning Platform"** as part of the curriculum of **Diploma in Computer Engineering**. Being novice in the field of designing and structuring in this micro-project, it could have been extremely difficult for us to complete this micro-project on our own. We are very fortunate to be guided by people with vast and resourceful experience in their respective field of work.

We express our sincere gratitude to our guide **Smt.Ajita.S.Patil(Lecturer, Computer Engg. Dept.).** For her timely guidance, support and suggestions. We arealso thankful for her sincere help and for making us available all the facilities of the department. Without her efforts and constant monitoring the micro-project and documentation would not have been duly completed. Also, we express our sincere thanks to **Mrs.Megha.G.Yawalkar (HOD Computer Department)**, besides, we take thisopportunity to express our sincere gratitude to the **Principal Dr. Vitthal.S.Bandal, G .P. Awasari** for providing a good environment and facilities to complete this micro-project. We would also like to thank all my colleagues who have directly or indirectly guided and helped us in the preparation of this micro-project.

Kadambari Nagare

Aditi Jadhav

Aditya Dhanwade

INDEX

Sr.No.	Contents	Page Number
1	Aim/Benefits of Microproject	1
2	Course Outcome Addressed	1
3	Proposed Methodology	1
4	Action Plan	2
5	Resources Required	3
6	Name of team members with enrollment	3

PART A

Micro-Project Proposal
on
E-Learning Platform

1.0 Aim/Objectives of the Micro-Project:-

The aim of the e-learning platform is to replace traditional classroom methods with accessible and interactive digital content, enhancing student engagement and learning outcomes. It ensures a user-friendly experience with diverse content formats and effective assessment tools.

2.0 Course Outcomes Addressed:-

CO1 : Develop programs using GUI Framework (AWT and Swing)

CO2 : Handle events of AWT and Swings components.

CO3 : Develop programs to handle events in Java Programming.

CO4 : Develop programs using database.

3.0 Proposed Methodology:-

The e-learning platform involved several key steps. Initially, a needs analysis was conducted to understand the educational requirements and preferences of the students, ensuring the platform addressed their needs effectively. Following this, concise and clear educational content was developed, focusing on text-based explanations and using examples to simplify complex topics.

The platform was designed with a user-friendly interface to facilitate easy navigation for students. Robust technologies were integrated to support the platform's performance and scalability, ensuring it was compatible with various devices and operating systems. Once ready, the platform was introduced in phases to ensure a smooth rollout, accompanied by training sessions and resources to help students familiarize themselves with the platform. Finally, the platform underwent continuous improvement based on technological advancements to remain relevant and effective.

4.0 Action plan:-

Sr. No	Details of activity	Planned starts date	Planned finish date	Name of responsible Team Members
1	Formation of project group.	22/07/2024	24/07/2024	1)Kadambari Nagare 2)Aditi Jadhav 3)Aditya Dhanwade
2	Allocation of project title by subject teacher.	31/07/2024	07/07/2024	
3	Conduct the information search about the project for requirement analysis of the project.	14/08/2024	21/08/2024	
4	Procurement/Arrangement of components and material required for project.	28/08/2024	04/09/2024	
5	Actual projector assembling work	18/09/2024	25/09/2024	
6	Testing calibration and proto-type development.	09/10/2024	16/10/2024	
7	Report preparation.	06/10/2024	30/10/2024	
8	Submission of project.	23/11/2024	15/11/2024	

5.0 Required Resources:-

Sr.no	Name of resources/materials	Specification	Qty	Remarks
1	Operating system	Windows 11	1	-
2	Development Software	Notepad/Netbeans	1	-
3	Notepad++/Notepad	-	1	-
4	Laptop	HP	1	-
5	Processor	I5 11th Generation	1	-
6	RAM	8 GB	1	-

6.0 Name of Team Members with Enrollment No:-

Sr No	Roll No	Name Of Student	Enrollment No
1	23CO349	Kadambari Hemant Nagare	2210510059
2	23CO350	Aditi Vijay Jadhav	2210510060
3	23CO351	Aditya Babasaheb Dhanwade	2210510061

INDEX

Sr.No.	Contents	Page Number
1	Rational	1
2	Aim of microproject	1
3	Course Outcome Integrated	1
4	Literature review	1
5	Actual methodology followed	2
6	Actual resource used	3
7	Skill Developed / Learning outcome of this Micro-Project	13
8	Applications of this Micro-Project	13
9	Conclusion	14
10	Reference	14

Part – B Micro-Project Report
On
E-Learning Platform

1] RATIONALE

The rationale for the E-learning platform project is to provide students with an interactive, accessible, and comprehensive resource for learning subjects like Physics, Chemistry, and Mathematics. In an age of digital transformation, traditional educational methods can be complemented with technological innovations that allow students to learn at their own pace, regardless of geographical limitations. This platform will serve as a repository for solutions, exercises, quizzes, and interactive content, enabling learners to engage deeply with academic subjects and build a strong foundation for their future studies.

2] AIM OF THE MICROPROJECT

The aim of the e-learning platform is to replace traditional classroom methods with accessible and interactive digital content, enhancing student engagement and learning outcomes. It ensures a user-friendly experience with diverse content formats and effective assessment tools.

3] COURSE OUTCOMES INTEGRATED

CO1 : Develop programs using GUI Framework (AWT and Swing)

CO2 : Handle events of AWT and Swings components.

CO3 : Develop programs to handle events in Java Programming.

CO4 : Develop programs using database.

4] LITERATURE REVIEW

I) 2021 IJCRT | Volume 9, Issue 4 April 2021 | ISSN: 2320-2882 :

In this paper, we explore how e-learning utilizes the internet and a combination of applications, termed as "platforms." These platforms, built with programming languages, function as software that relies on hardware for physical interaction with users. The paper seeks to define what e-learning truly means, given its increasing popularity and the growing number of users engaging with it daily.

II)The empirical literature vol. 115, no. 3, p. 1. M. (2013) :

In this journal, a meta-analysis synthesizes studies comparing learning outcomes for fully online and blended learning approaches, with an even split between college students and younger learners. The analysis offers insights into the effectiveness of these formats and emphasizes the need to tailor learning environments to suit different educational levels.

III)Problem Statement :

The platform being developed addresses these shortcomings by offering a modern e-learning experience. It provides personalized learning paths for students, interactive quizzes to reinforce learning, and tailored content that evolves based on student progress. With a user-friendly interface, the platform offers structured resources in Physics, Chemistry, and Mathematics that are designed for easy access. It integrates multimedia, gamified quizzes, and rich content to create an interactive, flexible learning environment.

IV) Disadvantage of existing system :

- Lack of Personalized Learning
- Limited Interaction and Feedback
- Poor User Experience
- Resource Gaps

V) Benefits of Proposed system :

- Personalized Learning Paths
- Interactive Learning
- Enhanced User Experience
- Comprehensive Resources

5] Actual Methodology Followed:

The e-learning platform involved a systematic approach that incorporated both technical and educational frameworks. The project was developed using advanced Java, integrated with MySQL for data management. The platform was divided into modules based on subjects like Physics, Chemistry, and Mathematics, and each module catered to specific grade levels (e.g., 1st to 5th standard). Key features, such as interactive quizzes, chapter-wise solutions, and progress tracking, were identified to design an intuitive user interface for students.

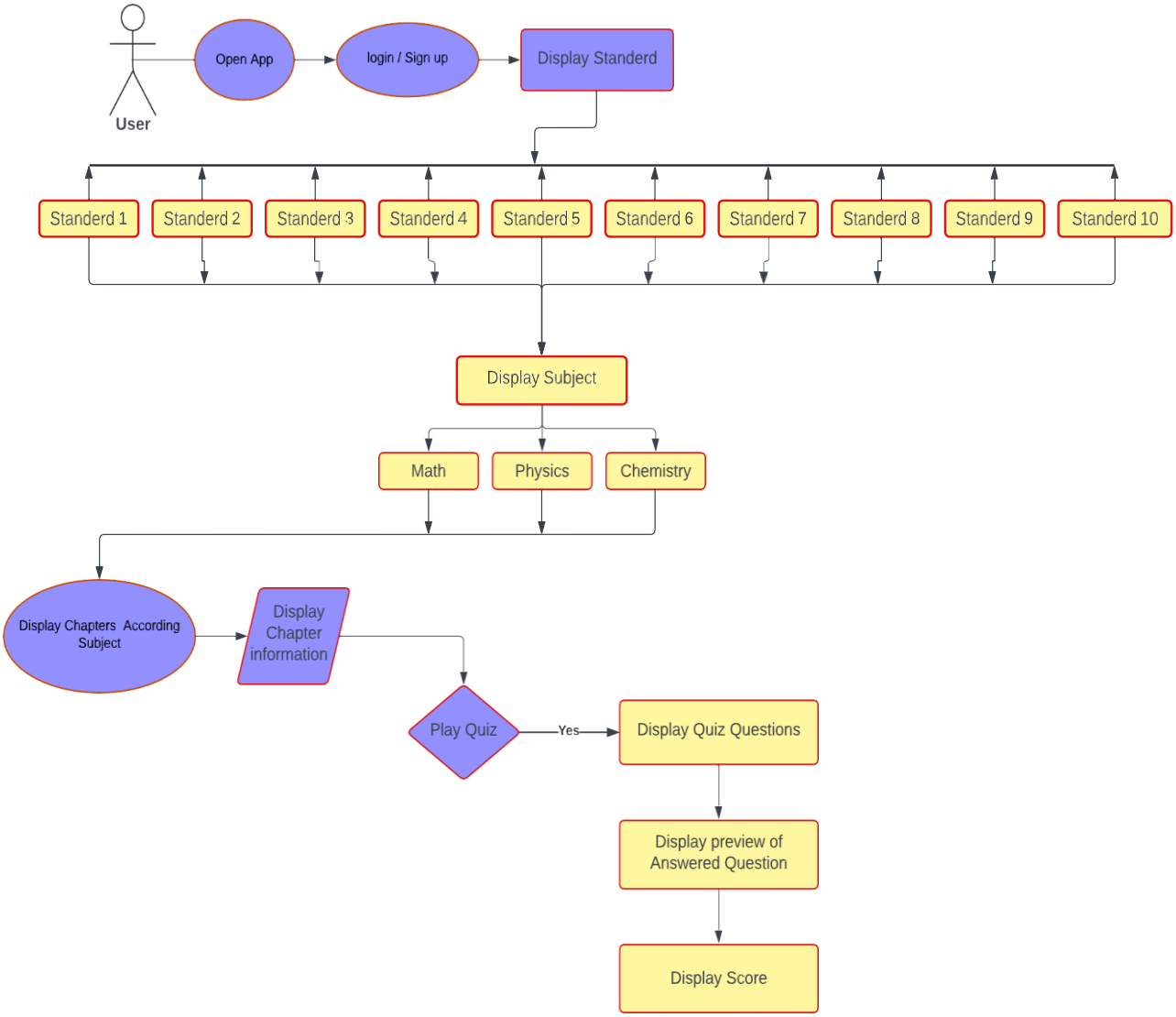
Database connections were used to store student performance, quiz results, and user details. Implementation was done using Java for backend logic, integrated with MySQL to store questions, user data, and progress. Chapters and quizzes were designed to adapt to user selections (e.g., standard, subject).

6] Actual Resources Used:

Sr.No.	Name of resource/ material	Specifications	Qty	Remarks
1.	Microsoft Edge	Execute the program	1	
2.	word	Designing the project information and typing	1	
3.	Software	Notepad ++, Eclipse IED	1	
4.	Database	MySQL	1	
5.	Computer System	Hp Laptop Intel Core 15 Ram: 8GB SSD:512 GB	1	
6.	Printer	Hp	1	

❖ Introduction :

- **Data-Flow Diagram :**



❖ Modules :-

1. User Profile and Progress Tracking Module:

This module allows students to create personalized profiles where their learning progress is tracked. Students can monitor completed topics, quizzes, and assessments for each subject.

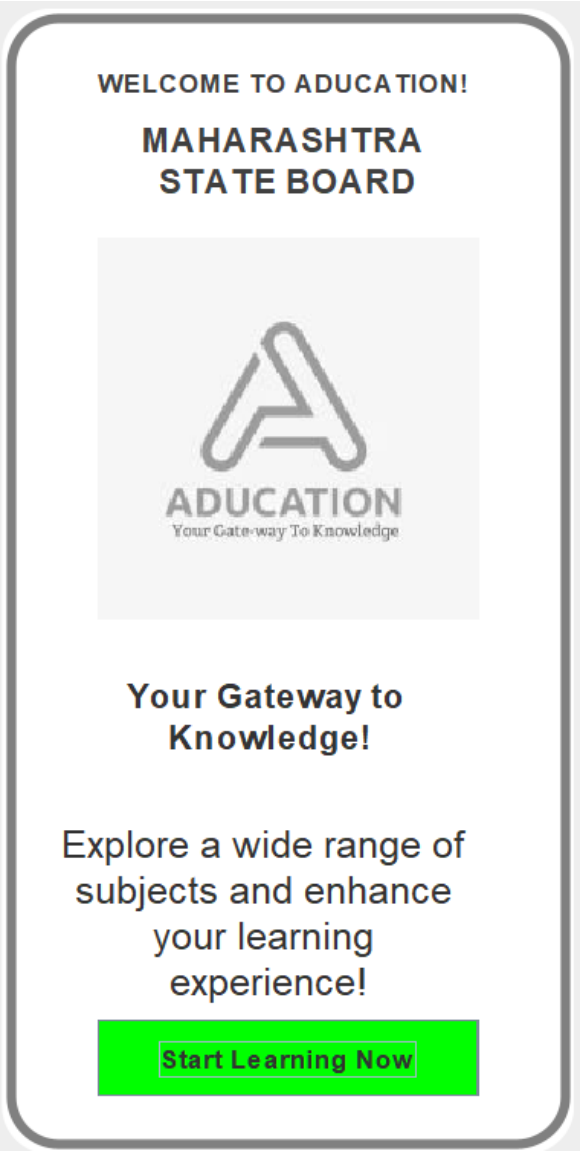
2. Interactive Learning Content Module:

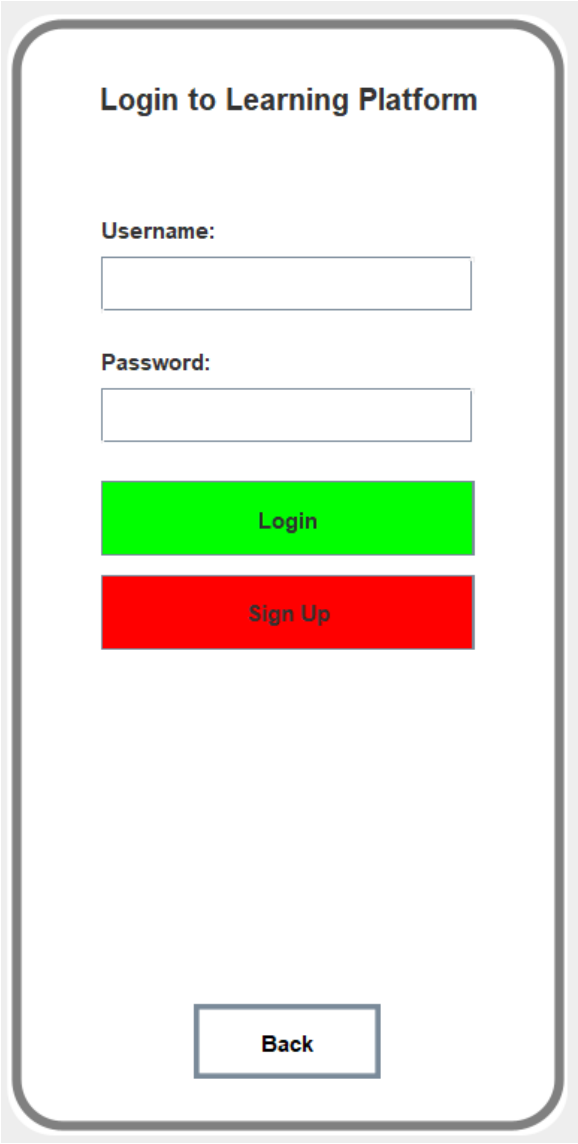
This module includes comprehensive and interactive learning materials for Physics, Chemistry, and Mathematics. Each subject is divided into chapters and lessons.

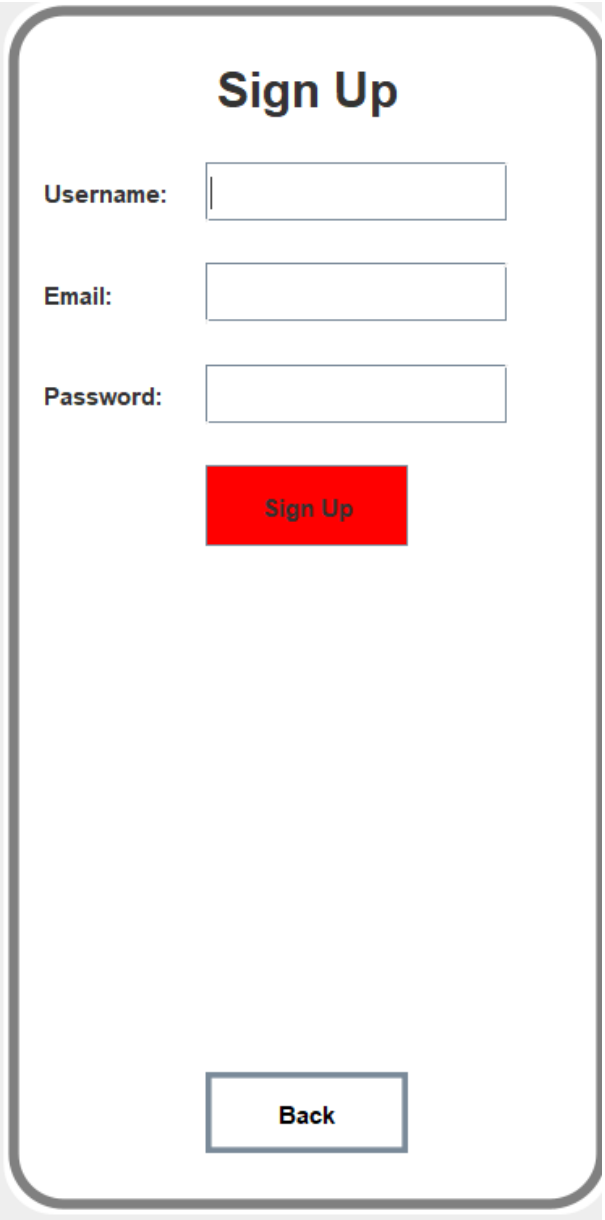
3. Quiz and Assessment Module:

This module offers quizzes, tests, and interactive assessments for each chapter. The questions are aligned with the curriculum and help reinforce learning. Immediate feedback is provided on quizzes, allowing students to learn from their mistakes.

❖ **Output :**

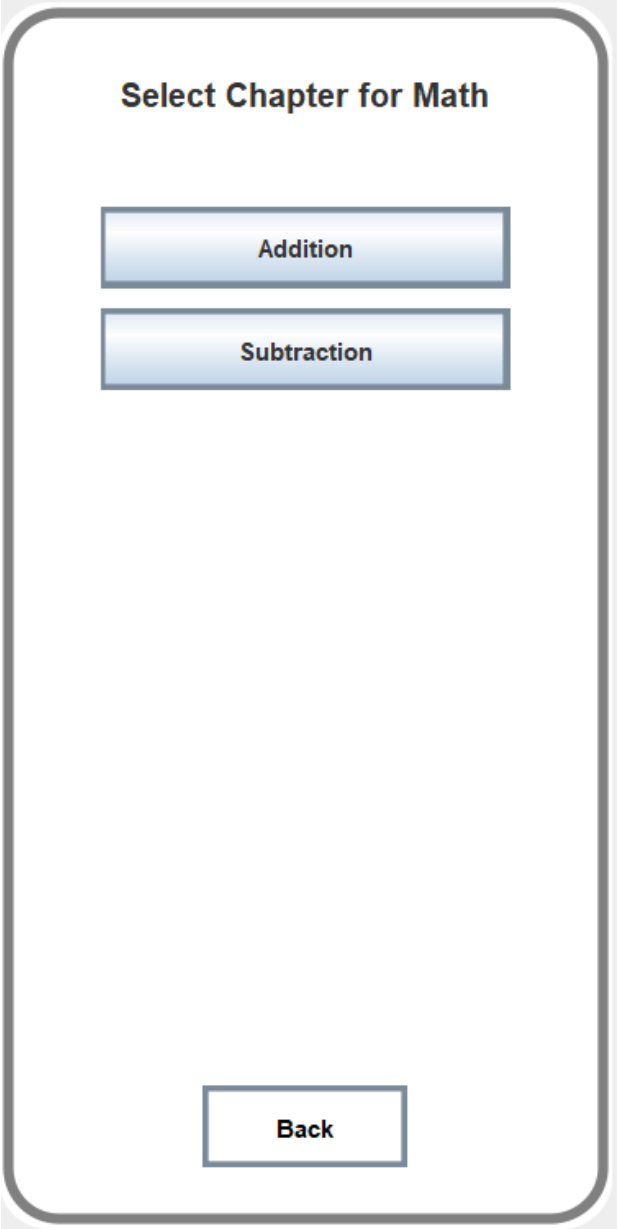

WELCOME TO ADUCATION!
MAHARASHTRA STATE BOARD
ADUCATION
Your Gate-way To Knowledge
Your Gateway to Knowledge!
Explore a wide range of subjects and enhance your learning experience!
Start Learning Now

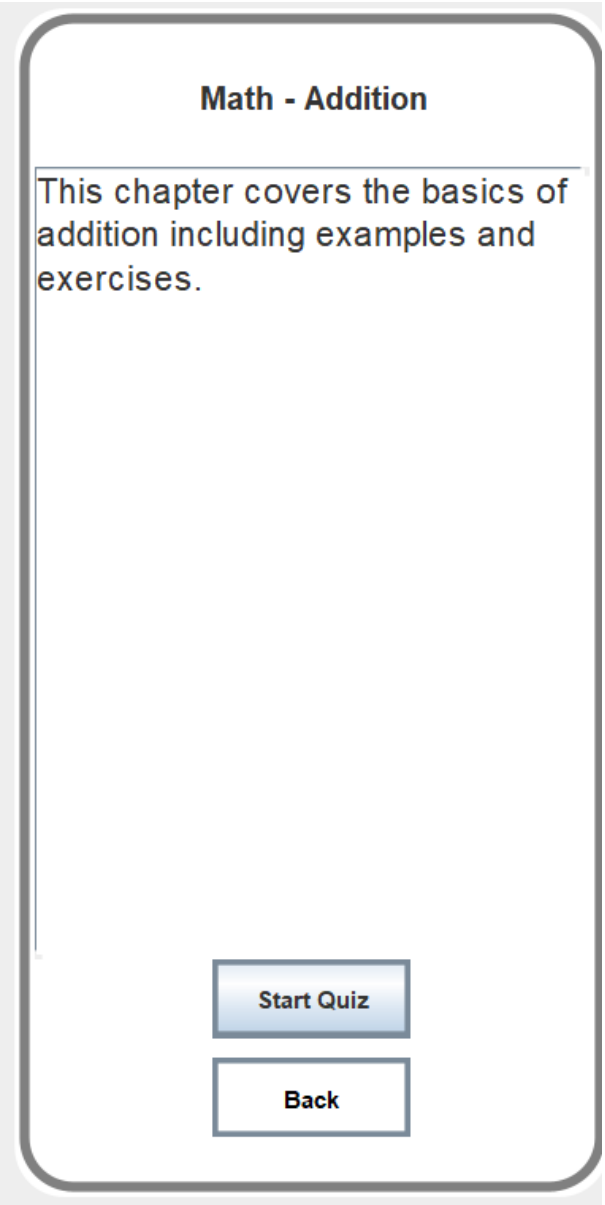

Login to Learning Platform
Username:
Password:
Login
Sign Up
Back

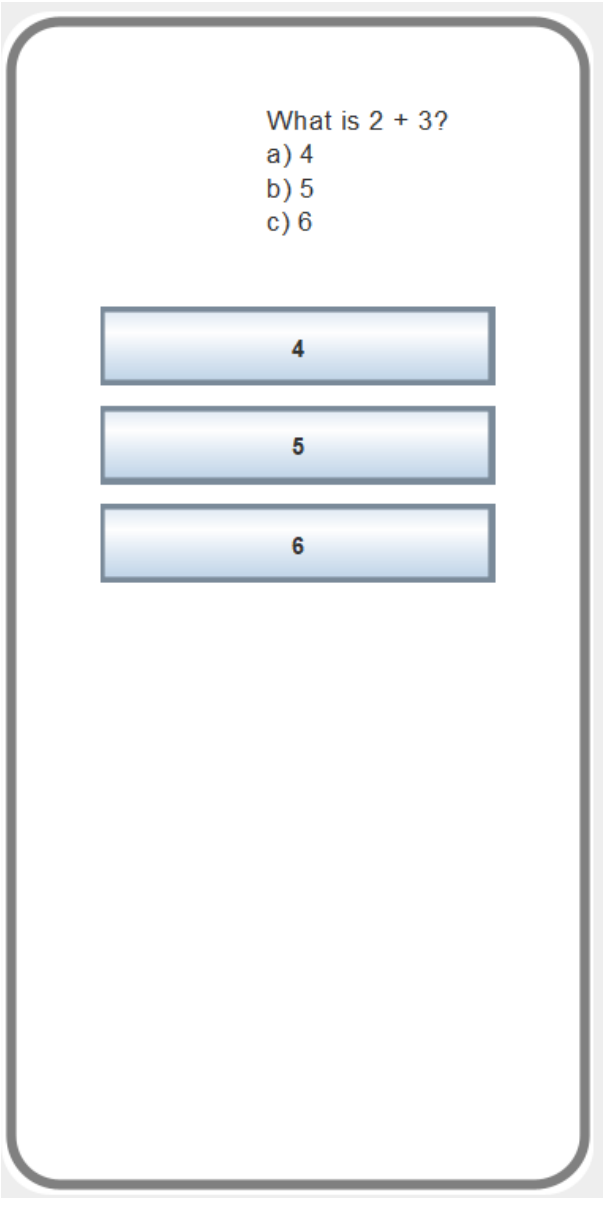

Sign Up
Username:
Email:
Password:
Sign Up
Back

Select Your Standard
Standard 1
Standard 2
Standard 3
Standard 4
Standard 5
Standard 6
Standard 7
Standard 8
Standard 9
Standard 10
Back

Select Subject
Math
Science
Back


Select Chapter for Math
Addition
Subtraction
Back


Math - Addition
This chapter covers the basics of addition including examples and exercises.
Start Quiz
Back


What is 2 + 3?
a) 4
b) 5
c) 6
4
5
6

Review your ans...

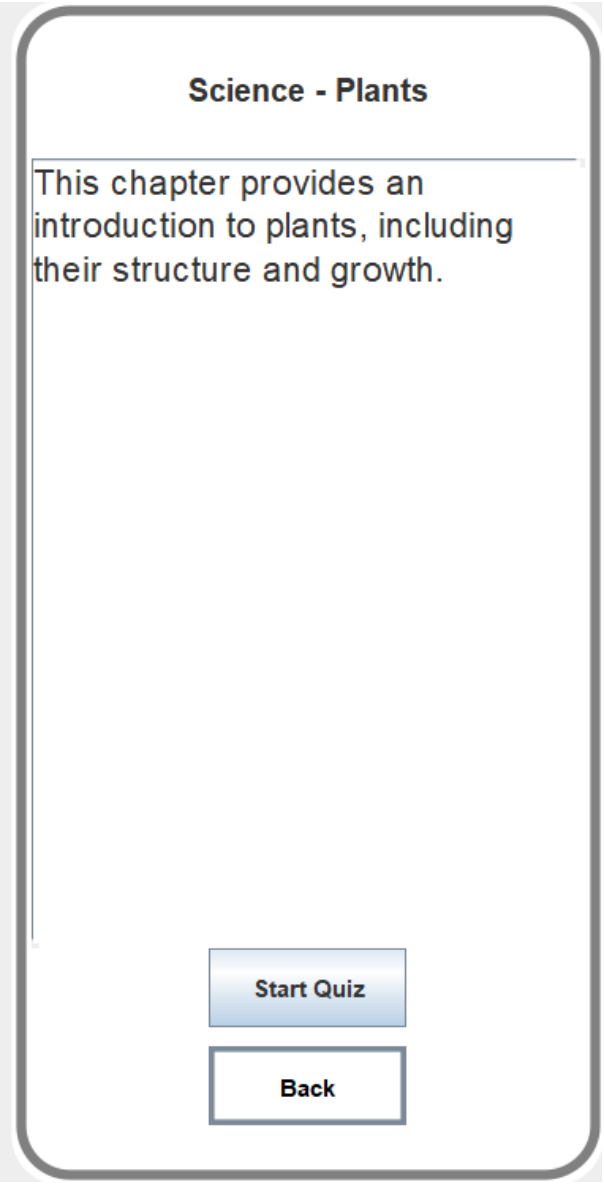
Q1: <html>What is 2 + 3?
a) 4
b) 5
c) 6</html>
a) 4
b) 5
c) 6
Your answer: 4
Q2: <html>What is 5 + 7?
a) 11
b) 12
c) 13</html>
a) 11
b) 12
c) 13
Your answer: 11
Q3: <html>What is 8 + 2?
a) 9
b) 10
c) 11</html>
a) 9
b) 10
c) 11
Your answer: 10

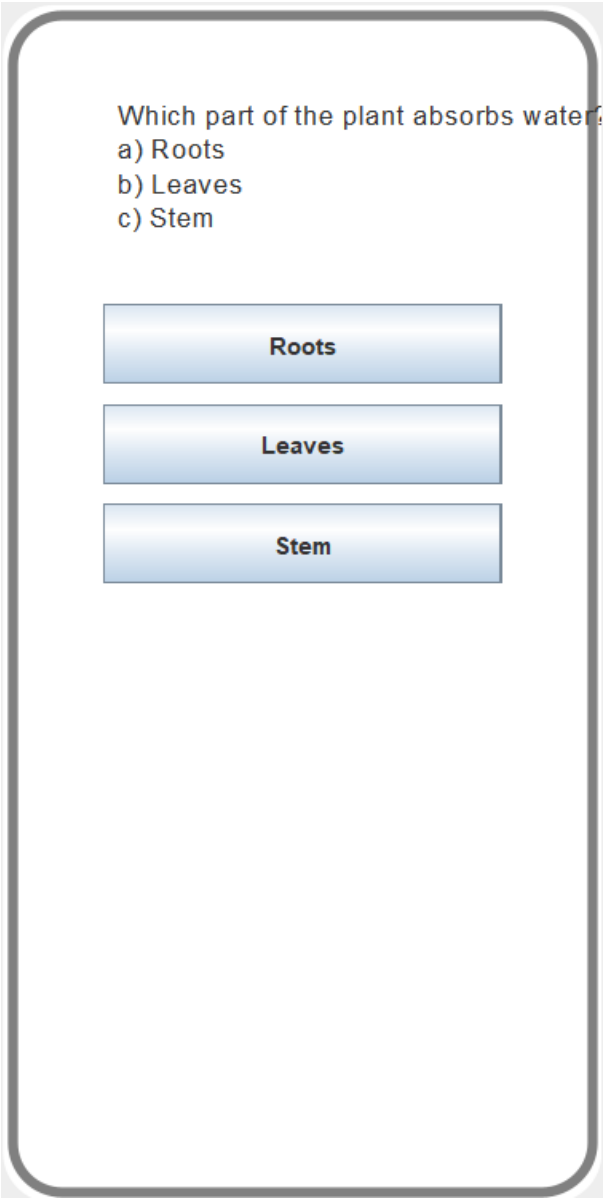
Select Chapter for Science

Plants

Animals

Back


Science - Plants
This chapter provides an introduction to plants, including their structure and growth.
Start Quiz
Back


Which part of the plant absorbs water?
a) Roots
b) Leaves
c) Stem
Roots
Leaves
Stem

Review your ans...

Q1: <html>Which part of the plant absorbs water?
a) Roots
b) Leaves
c) Stem</html>
a) Roots
b) Leaves
c) Stem
Your answer: Roots
Q2: <html>What is the process called where plants make food?
a) Photosynthesis
b) Transpiration
c) Respiration</html>
a) Photosynthesis
b) Transpiration
c) Respiration
Your answer: Transpiration

Submit Q... | Back to Q...

You scored: 1 out of 3

Back to Chapters

```
SELECT * FROM learning_platform.users;
```

id	username	email	password
1	aditya	aditya@gmail.com	123456
2	mahesh	mahesh123@gmail.com	987654
3	aditi jadhav	aditi@gmail.com	123456
4	kadamabari	kadamabarinagare45@gmail.com	12345

Schema: learning_platform

```
SELECT * FROM learning_platform.subjects;
```

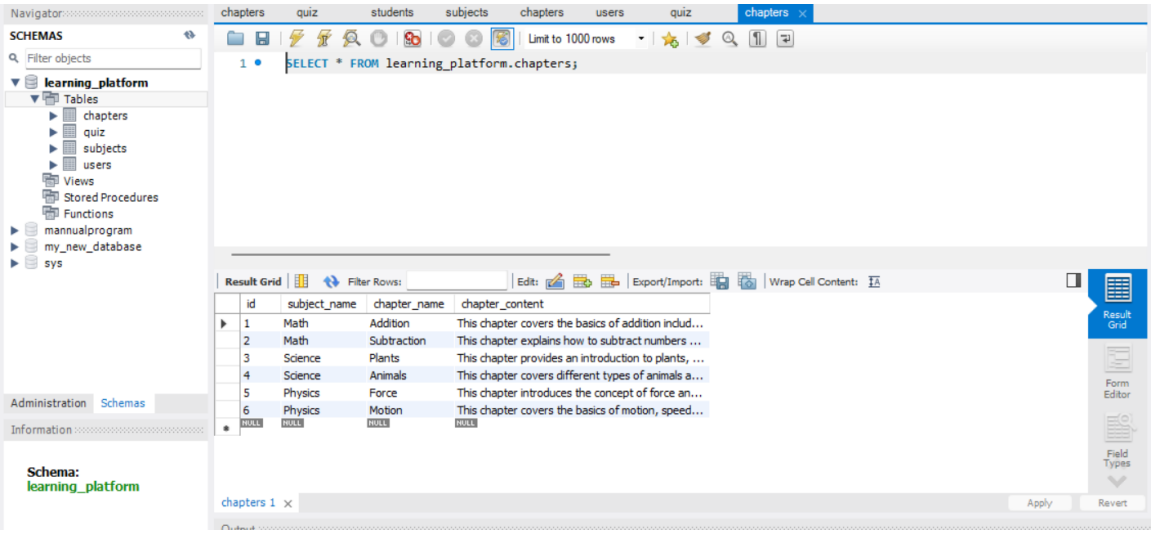
id	standard	subject_name
1	1	Math
2	1	Science
3	2	Math
4	2	Science
5	3	Math
6	3	Science
7	4	Math
8	4	Physics
9	5	Math
10	5	Physics

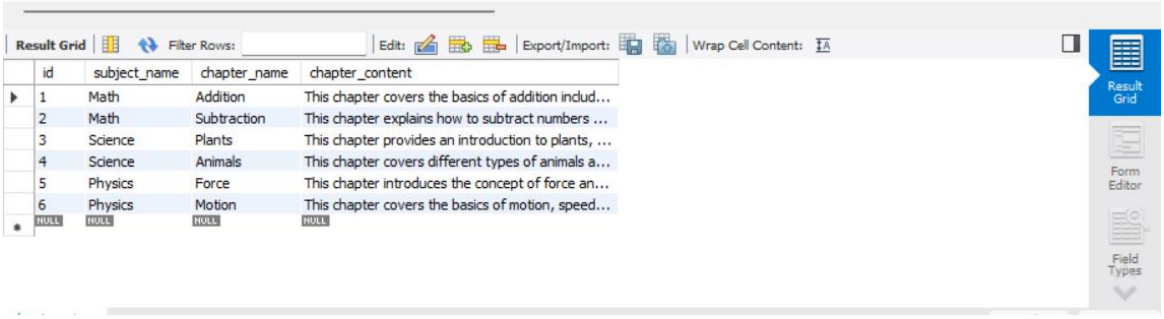
Schema: learning_platform

```
SELECT * FROM learning_platform.quiz;
```

id	chapter_name	question	option_a	option_b	option_c	correct_answer
1	Addition	What is 2 + 3?	4	5	6	b
2	Addition	What is 5 + 7?	11	12	13	b
3	Addition	What is 8 + 2?	9	10	11	b
4	Subtraction	What is 5 - 3?	1	2	3	b
5	Subtraction	What is 9 - 4?	4	5	6	b
6	Subtraction	What is 10 - 7?	2	3	4	b
7	Plants	Which part of the plant absorbs water?	Roots	Leaves	Stem	a
8	Plants	What is the process called where plants make f...	Photosynthesis	Transpiration	Respiration	a
9	Animals	Which animal is a mammal?	Frog	Cow	Lizard	b
10	Animals	What do herbivores eat?	Meat	Plants	Insects	b
11	Force	Which is a type of force?	Gravity	Heat	Light	a
12	Force	What unit is force measured in?	Newton	Watt	Volt	a
13	Motion	What is the rate of change of position called?	Speed	Force	Mass	a
14	Motion	What is the formula for speed?	Distance/Time	Mass/Volume	Force/Area	a

Schema: learning_platform





7] Skill Developed / Learning outcome of this Micro-Project :

- **UI/UX Design Skills:** to design user-friendly interfaces that enhance the learning experience for young students.
- **Database Management:** Gained experience in managing and organizing large sets of data,.
- **Java Programming Expertise :** Improved proficiency in Java, using advanced concepts to build an interactive and responsive platform.
- **Improved Project Management:** Gained experience in organizing the development process, including defining project scope, creating timelines, and coordinating tasks across different modules.

8] Applications of this Micro-Project:

- **Educational Institutions**: Schools and tutoring center's can adopt the platform to complement traditional teaching methods, providing students with additional resources and assessments.
- **Homeschooling**: The platform can serve as a valuable resource for homeschooling parents, offering structured content and assessment tools that facilitate independent
- **Content Creators**: Educators and content creators can use the platform to publish and share educational materials, quizzes, and resources tailored to various learning objectives.

9] Conclusion :-

This micro-project has led to the creation of an innovative e-learning platform designed for students in grades 1 to 5, offering engaging content, interactive assessments, and learning support. The project enhanced key skills such as technical development, project management, and user experience design. With room for future growth, the platform has the potential to expand its features and subjects, contributing to the advancement of digital education.

10] Reference:

➢ **Books :**

- Complete Reference , Schildt, Herbert
- Java 2 Programming Black Book ,Holzer, Steven et al

➢ **Journal Papers :**

1. International Journal of Emerging Technologies in Learning (iJET). 2016, Vol. 11 no. 1, pp. 11- 20, 10p, Feb. 2016.

2. International Journal of Creative research thought 2021 IJCRT | Volume 9, Issue 4 April 2021 | ISSN: 2320-2882.

➢ **Website :**

1.https://in.search.yahoo.com/search?fr=mcafee&type=E211IN714G0&p=lucid+flow+chart&guccounter=1

2.https://elearningindustry.com/directory/software-categories/learning-management-systems